Ian Q. Snider

+1 (660) 341-6806 · Berkeley, CA · iqsnider@berkeley.edu · iansnider.com · github.com/iqsnider

EDUCATION

University of California, Berkeley, *Berkeley, CA* — *Aug 2025 - present*
Graduate Researcher in Nuclear Engineering — --/4.00

Washington University in St. Louis, *St. Louis, MO* — *May 2025*
B.S. Mechanical Engineering — 4.00/4.00

Truman State University, *Kirksville, MO* — *Conferred: Dec 2024*
B.A. Physics, Mathematics Minor — 3.89/4.00

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, Berkeley, CA — *Aug 2025 - present*

- Graduate researcher in the Bay Area Neutron Group
- Researching experimental techniques for indirect neutron capture cross-section measurement of short-lived nuclei

Lawrence Livermore National Laboratory - *Graduate Intern*, Livermore, CA — *June 2025 - present*

- Developed CRISP (CRItical Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP sucessfully automated the materials compiling and k_{eff} benchmarking of ~3420 ICSBEP critical reactors

Brookhaven National Laboratory - *Nuclear Science Intern*, Upton, NY — *Summers 2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for heavy-nuclei resonance reclassification on NNDC clusters
- Applied Python machine learning using Scikit-learn and Pytorch to classify resonance spin assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used nuclear data and Random Matrix Theory to develop synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

Truman State University - *Student Astronomy Researcher*, Kirksville, MO — *2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI
- Researched long-exposure luminosity data corruption due to satellite interference

SKILLS

- **Programming:** Python, Go, C, C++, Bash, Lua, SQL, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino
- **Physics/engineering:** Electrodynamics, Electronics, Classical Mechanics, Quantum Physics, Mathematical Physics, Vibrations, Nuclear Physics, Thermodynamics, Fluid Mechanics, Solid Mechanics, Heat Transfer, Acoustics, Materials Science, Thermal Systems, Turbojets, Ramjets, Autonomous Aerial Vehicle Control
- **Mathematics:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, and Optimizations

ACTIVITIES

WashU Robotics MATE ROV & MARINER Projects - *Mechanical Lead* — *2023 - 2025*

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Prototyped hydrodynamic dive control using an IMU and sonar state estimation with a Kalman Filter
- Built a chassis and buoyancy engine

Society of Physics Students - *Demo Chair* — *2020 - 2023*

- Developed 3 new physics demos, performed demos at meetings, and provided volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads (“Crave the Wave” and “Remote Sensing”)